FORGE PRESENTS

TapGuard IQ

Keep Your Tap. Make It Smart.

Smart retrofit system for reducing avoidable water wastage

The Problem

Taps are sometimes left running accidentally, especially in busy or shared places. A few unnoticed minutes can waste several litres of water before somebody notices and closes the tap.

The Innovation

TapGuard IQ is a proposed retrofit system that adds intelligence to an existing tap instead of replacing it. It combines water-flow sensing, mmWave human-presence detection, time and the selected operating mode to identify possible unattended water flow. When necessary, it can warn the user and control an electronic valve to stop the water.

WATER IN	FLOW SENSOR	AUTO VALVE	EXISTING TAP
	measures flow	controls water	normal tap remains

Three Proposed Modes

Normal Mode	If water keeps flowing after the user leaves, the system can warn and automatically stop the flow.
Fill Mode	The user selects a quantity such as 500 mL or 1 litre. The system measures the water and stops at the selected target.
Override Mode	Allows intentional continuous use for a limited time when the user knows the water needs to keep running.

What Makes It Different?

A basic sensor tap mainly reacts to immediate presence. TapGuard IQ is designed to consider **flow + presence + time + user intention**. This matters because walking away does not always mean water is being wasted - for example, someone may intentionally be filling a container. The aim is to distinguish these situations while keeping the existing tap.

Where It Can Be Used

Schools, restaurants and commercial kitchens, hostels, hospitals, offices, public wash areas, hotels, apartment buildings and homes - especially locations where many people share the same water points.

Potential Impact

A running tap can use several litres of water every minute. By stopping accidental unattended flow earlier, TapGuard IQ could prevent avoidable losses. The working prototype will measure actual flow so that water-saving results can be calculated from real tests rather than estimated claims.

Proposed Prototype

ESP32 controller + water-flow sensor + mmWave presence sensor + normally-closed electronic valve + simple display/controls. The first prototype will use a small low-voltage water setup for safe testing and demonstration.